

Number Theory Toolkit

Primes, GCD, Modular Arithmetic · Mentor-Led Lab

Developed by TalenciaGlobal

Difficulty ★★★ Intermediate

Problem

Given three positive integers a , b and m , build a small number-theory toolkit that reports several classic results about them.

What to Compute

- Whether a is prime (print Is Prime = true or false).
- The greatest common divisor $\text{GCD}(a, b)$.
- The least common multiple $\text{LCM}(a, b) = a / \text{GCD}(a, b) * b$.
- The number of primes from 1 to a (use a sieve).
- $(a * b) \bmod m$ and $(a + b) \bmod m$.

Input / Output Format

Input: Three integers a b m .

Output: Six labelled lines: Is Prime, GCD, LCM, Primes up to a , $(a * b) \bmod m$, $(a + b) \bmod m$.

Examples

Example 1

Input: $a = 12$, $b = 18$, $m = 7$

Output:

```
Is Prime = false
GCD = 6
LCM = 36
Primes up to a = 5
(a * b) mod m = 6
(a + b) mod m = 2
```

Explanation: 12 is not prime; $\text{gcd}(12, 18) = 6$; $\text{lcm} = 12/6 * 18 = 36$; the primes up to 12 are 2, 3, 5, 7, 11 (five of them); $216 \bmod 7 = 6$; $30 \bmod 7 = 2$.

Example 2

Input: $a = 13$, $b = 39$, $m = 5$

Output:

```
Is Prime = true
GCD = 13
LCM = 39
Primes up to a = 6
(a * b) mod m = 2
(a + b) mod m = 2
```

Explanation: 13 is prime; gcd = 13; lcm = 39; there are six primes up to 13; $507 \bmod 5 = 2$; $52 \bmod 5 = 2$.

Constraints

- $1 \leq a, b \leq 10^6$
- $2 \leq m \leq 10^9$
- Compute LCM as $a / \text{GCD}(a, b) * b$ to avoid overflow.

Approach

Primality by trial division up to $\sqrt{a}$: if any i from 2 to $\sqrt{a}$ divides a , it is not prime.

GCD by the Euclidean algorithm: repeatedly replace (a, b) with $(b, a \bmod b)$ until b is 0; the last non-zero value is the GCD.

LCM from the identity $a * b = \text{GCD} * \text{LCM}$. Divide before multiplying $(a / g * b)$ so the intermediate value stays small.

Prime count with the Sieve of Eratosthenes up to a , then count the survivors.

Modular arithmetic is just the $\%$ operator: $(a + b) \bmod m$ and $(a * b) \bmod m$ follow directly.

Sample Run

Sample Input

```
12 18 7
```

Sample Output

```
Is Prime = false
GCD = 6
LCM = 36
Primes up to a = 5
(a * b) mod m = 6
(a + b) mod m = 2
```